UNIT 1 · KINEMATICS · WEEK 3 · LAB DAY 1 OF 2

Cart Lab — Day 1: Run & Record

Physics 2026–27 · in class Thursday, October 8 · bring this sheet back Tuesday

NAME _____ GROUP _____

The question: what is the relationship between position and time for a cart accelerating from rest down a ramp? Today you build, predict, run, record, and plot. Tuesday you read the acceleration off this data.

1 · Build the ramp

CARDBOARD · 15 MIN

- **Straight.** A ramp that sags changes slope partway down, and then the acceleration is not constant. Fold the cardboard into a channel or support it underneath so a meter stick lies flat along it.
- **Shallow.** Raise the top end only a few centimetres. The cart should take **at least 2 seconds** to roll the full length; if it is faster, lower the ramp.
- **Fixed.** Tape both ends down, add side rails so the cart rolls straight, and put a backstop at the bottom. Once you start timing, the ramp does not move.

Measure and record: ramp length _____ m height of the top end _____ m what is rolling _____

2 · Predict before your first timed run

HYPOTHESIS · 5 MIN

a. Released from rest, does the cart cover more distance in the **first half** of its travel time or the **second half**? Circle one. Because: _____

b. If a is constant, $x = \frac{1}{2}at^2$. So rolling **four times as far** should take _____ times as long.

c. Sketch position vs time in the box. Label both axes with units.

3 · Set up

DECIDE ONCE, THEN STAY CONSISTENT · 5 MIN

- **Origin at the release line; down the ramp is positive.** Positions in metres, by meter stick.
- **Front of the cart on the release line** every time. Let go — do not push.
- **Five evenly spaced tape marks**, measured from the release line: `0.20, 0.40, 0.60, 0.80, 1.00 m` if your ramp is long enough. On a shorter ramp use a smaller step, such as `0.15, 0.30, 0.45, 0.60, 0.75 m`.
- Each run times **release → one mark**. Switch jobs after every mark.

RELEASER _____ TIMER _____ RECORDER _____

4 · Data

THREE TRIALS PER MARK · 25 MIN

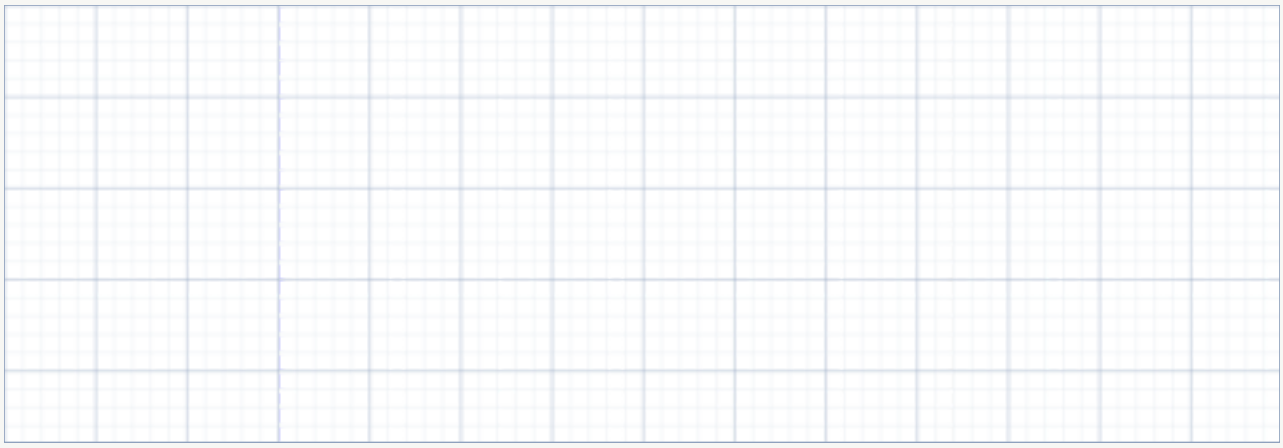
MARK	POSITION X (M)	TRIAL 1 T (S)	TRIAL 2 T (S)	TRIAL 3 T (S)	AVERAGE T (S)
1					
2					
3					
4					
5					

Table rules: numbers only in cells; times to the hundredth (`1.40`, not `1.4`); cross out a bad trial and rerun.

5 · Plot your points

BEFORE YOU PACK UP · 10 MIN

Plot **average time** across and **position** up. Label both axes with the variable and its unit, choose a scale that uses most of the grid, and include the point `(0, 0)`. **Dots only — no line yet.**



6 · Check before you leave

EXIT CHECK · 5 MIN

a. Do your points curve upward, or do they fall on a straight line? If straight, tell me before you take the ramp apart.

b. Divide your average time at **mark 4** by your average time at **mark 1**: _____. Compare with your prediction in 2(b).

c. Which mark had the biggest spread between its three trials? Why was it the hardest to time?

ANSWERS

Keep this sheet *and photograph the table and your ramp. Lab Report I is due Thu 10/15.*